Moamen Elsharkawy

Applied AI Engineer | AI Agents, RAG and Workflow Automation

Cairo, Egypt | +20 100 097 7841 | contact@moamen-ai.com

GitHub | Portfolio | LinkedIn

Summary

Hands-on AI engineer building tool-using agents, retrieval systems and integrations with Python, TypeScript and n8n. Implemented tailored AI Operating Systems used internally at two real-estate companies. Experience includes CRM integration, human approval, handling retries safely and testing failure paths.

Technical Skills

Applied AI: Large language models (LLMs), retrieval-augmented generation (RAG), vector search, structured outputs, tool calling, evaluation, human approval

Software: Python, TypeScript, FastAPI, Node.js, PostgreSQL, Git, Docker

Integration: n8n, REST APIs, webhooks, CRM, idempotency, retries, audit trails

Experience

Chief AI Officer (part-time) | Home Locator

Aug 2026 - Present | Cairo, Egypt

- Build and integrate new components around an existing CRM and website for customer management and sales operations. Own implementation and troubleshooting for workflows used internally.
- Work with the owner to define AI and technology priorities and turn operational requirements into a delivery roadmap.

Chief AI Officer (part-time) | Real Experts

Jul 2026 - Present | Cairo, Egypt

- Built the original Egyptian-Arabic AI Operating System and CRM used internally by the brokerage.
- Own the architecture and implementation, translating brokerage processes into operating rules, structured data and connected workflows.

Selected Engineering Projects

Lead Intelligence Pipeline | Tested engineering reference

- Implemented n8n lead ingestion and Odoo CRM routing with deterministic scoring, separate LLM classification, an idempotency ledger, retries and dead-letter replay.
- Passed 194 local assertions in Oct 2026. A historical deployed test run passed 15 of 15 cases.

Multimodal Document QA | Academic project

- Built a Python pipeline for visually rich PDFs with a ColPali page retrieval path, TF-IDF fallback, page citations and a Streamlit interface.
- Implemented a benchmark harness that records ranked page references, answer keyword coverage and latency.

Additional Engineering Projects

Transcript Translator | Paid client delivery

- Delivered an n8n pipeline that accepts timestamped DOCX transcripts in Telegram, translates them into Arabic in chunks, restores segment order and returns a Drive link while preserving timestamps.

Sawt | Deployed showcase

- Built an Egyptian-Arabic voice agent that routes requests to business tools and asks for human confirmation before sensitive actions. Demonstrated it at n8n Cairo Meetup.

Chest X-ray Detection and Classification | Academic team project

- Integrated deep learning classifiers and object detectors into a unified Python inference pipeline for DICOM chest X-rays.
- Documented training, preprocessing and evaluation code, hyperparameter optimization and Grad-CAM explanations for classification.

ARES / Jezero Route Lab | Geospatial software prototype

- Built a browser terrain viewer using public HiRISE elevation data and compared A* route cost strategies. Published source, a demo and tests for the route and data pipeline.

Additional Experience

AI Automation Course Developer and Instructor (contract) | EYouth

Aug 2026 - Present

- Develop course material, scripts and n8n demos for AI Automation Mastery.

AI Automation Engineer and Instructor | Sparkling for Business LLC

May 2026 - Sep 2026 | Completed cohort work

- Built 13 validated teaching workflows, led office hours and supported student projects.

Instructor and Program Lead | Bonyan Academy

Aug 2025 - Apr 2026

- Taught AI and programming and coordinated instructors.

Full-Stack Development Intern | NTG Clarity

Jul 2025 - Sep 2025

- Built an e-learning application with Spring Boot, Angular and REST APIs.

Education

BSc in Artificial Intelligence and Data Science | Expected 2027

Zewail City of Science, Technology and Innovation, Egypt

Awards and Speaking

First-place team, n8n Hackathon Cairo (Nov 2026): co-built a social content workflow with research, drafting, image generation and human approvals. Speaker, n8n Cairo Meetup (Jul 2026).

Languages

Arabic | English